

Author Correction: CellFM: a large-scale foundation model pre-trained on transcriptomics of 100 million human cells

Correction to: *Nature Communications* <https://doi.org/10.1038/s41467-025-59926-5>, published online 20 May 2025.

In the original version of this article, we identified several errors in the performance results. Specifically, the average performance values in Fig. 4e, Fig. 4f, and Supplementary Fig. S15 were incorrectly calculated. Two baseline methods in Supplementary Fig. S19 were incorrectly evaluated on two datasets due to a bug in the evaluation script. After correction, the conclusions of this article remain unchanged (the outperformance of our method is mostly strengthened). Please find below the detailed corrections, where the original and corrected values are shown in purple and red, respectively.

1) Correction of average values in Fig.4e-4f for several baseline methods:

Average values for several baseline methods in Fig. 4e and Fig. 4f were incorrectly calculated. The table below shows the original and corrected average values for Fig. 4e-4f (corrected values are shown in red). Two scGPT result values in Fig. 4f were incorrectly updated during figure generation (0.367 to 0.347 and 0.255 to 0.248 on the Lung and Heart datasets, respectively). CellFM remains the best-performing method.

Fig4e									
Models	hPancreas	Liver	Pancrm	Lung	PBMC	Skin	Heart	Original Average	Corrected Average
scGPT	0.7626	0.52136	0.55568	0.62446	0.73636	0.8638	0.40086	0.6378743	0.6378743
GeneCompass	0.52734	0.55696	0.6934	0.6684	0.74644	0.87962	0.6288	0.6715657	0.6715657
Geneformer	0.94854	0.61976	0.82812	0.55882	0.6779	0.65068	0.40448	0.6697571	0.6697571
scmap	0.6785	0.8466	0.82742	0.5199	0.2375	0.7489	0.405	0.7310000	0.6091171
scELMo	0.97444	0.7366	0.9066	0.78632	0.82644	0.92942	0.70786	0.8382400	0.8382400
UCE	0.88468	0.8972	0.8412	0.89656	0.8479	0.97808	0.86358	0.8870286	0.8870286
scBERT	0.94392	0.84822	0.91798	0.8813	0.92262	0.97024	0.8818	0.9094400	0.9094400
scFoundation	0.96056	0.90434	0.91244	0.88708	0.89408	0.9775	0.9061	0.9203000	0.9203000
SVM	0.9687	0.92016	0.9193	0.9027	0.8957	0.9802	0.899	0.9230000	0.9265371
CellFM(800M)	0.9513	0.7294	0.8111	0.6549	0.8492	0.9371	0.8052	0.8197429	0.8197429
CellFM(80M)	0.97798	0.91234	0.9623	0.88524	0.97648	0.9798	0.9079	0.9431486	0.9431486
Fig4f									
Models	hPancreas	Liver	Pancrm	Lung	PBMC	Skin	Heart	Original Average	Corrected Average
scGPT	0.55862	0.2355	0.28878	0.3471	0.5392	0.53226	0.2482	0.3428000	0.3928086
GeneCompass	0.18526	0.29778	0.434	0.492	0.58172	0.55482	0.47274	0.4112000	0.4311886
Geneformer	0.86444	0.3052	0.70396	0.334	0.453886	0.56446	0.25842	0.4977666	0.4977666
scmap	0.6748	0.73258	0.71942	0.3334	0.2051	0.656	0.2658	0.6263000	0.5124429
scELMo	0.9178	0.6374	0.7976	0.62788	0.77622	0.83964	0.57112	0.7382371	0.7382371
UCE	0.7492	0.60394	0.55988	0.81554	0.791	0.92202	0.77366	0.7450343	0.7450343
scBERT	0.5839	0.72016	0.80256	0.7958	0.90146	0.9158	0.8148	0.7906400	0.7906400
scFoundation	0.74588	0.84058	0.84204	0.81614	0.87074	0.9376	0.84396	0.8424200	0.8424200
SVM	0.9229	0.83782	0.83064	0.7756	0.8632	0.9296	0.8115	0.8430371	0.8530371
CellFM(800M)	0.6383	0.5052	0.5919	0.5247	0.8218	0.8458	0.6804	0.6583000	0.6583000
CellFM(80M)	0.87424	0.79922	0.89116	0.7918	0.97486	0.9397	0.842	0.8732829	0.8732829

Correspondingly, the text will be updated from: “CellFM outperformed SVM, scmap, and scBERT on inter-dataset cell annotation tasks, with average accuracy improvements of 2%, 21.1%, and 3.4%, respectively.”

to: “CellFM outperformed SVM, scmap, and scBERT on inter-dataset cell annotation tasks, with average accuracy improvements of **1.66%, 33.4%, and 3.4%, respectively**”.

2) Correction of the average values in Supplementary Fig. S15:

The average values in Supplementary Fig. S15 were incorrectly calculated. The corrected values show only minor changes across the methods. CellFM remains the best-performing method.

1	Supplementary Fig. S15										
2	AVGBIO	hPancreas	Liver	Pancrm	Lung	PBMC	Skin	Heart	Original Average	Corrected Average	
3	UCE		0.7065	0.77622	0.65072	0.7706	0.6569	0.8901	0.6701	0.735773333	0.731591429
4	scBERT		0.8784	0.67204	0.84984	0.7412	0.6899	0.8824	0.6976	0.755496667	0.773054286
5	scFoundation		0.8598	0.80366	0.87606	0.7704	0.7152	0.8772	0.6817	0.797717143	0.797717143
6	CellFM		0.9056	0.8163	0.91414	0.8956	0.9459	0.9445	0.7498	0.881691429	0.881691429
7											
8	NMI	hPancreas	Liver	Pancrm	Lung	PBMC	Skin	Heart	Original Average		
9	UCE		0.7466	0.818	0.6621	0.8216	0.7073	0.9205	0.7559	0.756833333	0.776
10	scBERT		0.916	0.732	0.8801	0.793	0.7552	0.9317	0.7758	0.815493333	0.826257143
11	scFoundation		0.8934	0.8421	0.9102	0.8336	0.7392	0.9167	0.7621	0.860426667	0.842471429
12	CellFM		0.9217	0.8222	0.9345	0.9392	0.9699	0.9511	0.7971	0.907042857	0.9051
13											
14	ARI	hPancreas	Liver	Pancrm	Lung	PBMC	Skin	Heart	Original Average		
15	UCE		0.7259	0.7981	0.6585	0.7896	0.6223	0.9385	0.6118	0.7314	0.734957143
16	scBERT		0.9595	0.6603	0.9172	0.8037	0.6527	0.9587	0.7015	0.7976	0.807657143
17	scFoundation		0.9467	0.8397	0.941	0.8065	0.7434	0.9313	0.649	0.865353333	0.8368
18	CellFM		0.9563	0.8529	0.9518	0.9399	0.9701	0.9742	0.7638	0.918214286	0.915571429
19											
20	ASW	hPancreas	Liver	Pancrm	Lung	PBMC	Skin	Heart	Original Average		
21	UCE		0.647	0.7126	0.6315	0.7007	0.6412	0.8114	0.6426	0.67758	0.683857143
22	scBERT		0.7596	0.6238	0.7521	0.6268	0.6617	0.7569	0.6154	0.68664	0.685185714
23	scFoundation		0.7393	0.7293	0.777	0.6711	0.663	0.7836	0.634	0.734806667	0.7139
24	CellFM		0.8389	0.7738	0.8561	0.8077	0.8976	0.9082	0.6885	0.832933333	0.8244
25											
26	graph conn	hPancreas	Liver	Pancrm	Lung	PBMC	Skin	Heart	Original Average		
27	UCE		0.9159	0.9733	0.8969	0.9138	0.9583	0.9615	0.9827	0.938886667	0.9432
28	scBERT		0.9425	0.9272	0.9684	0.9298	0.9727	0.9675	0.9869	0.951806667	0.956428571
29	scFoundation		0.937	0.9704	0.9894	0.904	0.979	0.9696	0.9897	0.971873333	0.962728571
30	CellFM		0.9505	0.9766	0.9919	0.9761	0.9901	0.9835	0.981	0.979671429	0.978528571

3) Correction of results for two baseline methods in Supplementary Fig. S19:

In Supplementary Fig. S19, the results of UCE and scELMo on two datasets were affected by a bug in the evaluation script. Most corrected average scores decreased, except that the graph connectivity metric of UCE slightly increased. CellFM remains the best-performing method.

1	Supplementary Fig. S19						
2	avg_bio	scGPT	Original UCE	Corrected UCE	Original scELMo	Corrected scELMo	CellFM
3	HumanBrainCellAtlas	0.3823	0.4102	0.4102	0.2111	0.2111	0.4387
4	TabulaSapiens_All_Cells_v1	0.5553	0.5566	0.6025	0.4354	0.2459	0.5639
5	TabulaSapiens_All_Cells_v2	0.5706	0.5997	0.3412	0.4828	0.2386	0.6165
6	PBMC 10k	0.7820	0.7430	0.7430	0.6330	0.6330	0.7890
7	average	0.5726	0.5774	0.5242	0.4406	0.3321	0.6020
8							
9	ari						
10	HumanBrainCellAtlas	0.1082	0.1292	0.1292	0.0397	0.0397	0.1326
11	TabulaSapiens_All_Cells_v1	0.4541	0.4505	0.4973	0.4505	0.1227	0.4659
12	TabulaSapiens_All_Cells_v2	0.5450	0.5534	0.1081	0.5534	0.1219	0.5920
13	PBMC 10k	0.8464	0.8560	0.8560	0.6292	0.6292	0.9124
14	average	0.4884	0.4973	0.3977	0.4182	0.2284	0.5257
15							
16	nmi						
17	HumanBrainCellAtlas	0.4768	0.5048	0.5048	0.2819	0.2819	0.5021
18	TabulaSapiens_All_Cells_v1	0.6836	0.6827	0.7737	0.6827	0.4421	0.7494
19	TabulaSapiens_All_Cells_v2	0.7110	0.7115	0.5138	0.7115	0.4102	0.7529
20	PBMC 10k	0.8163	0.8210	0.8210	0.6983	0.6983	0.8702
21	average	0.6719	0.6800	0.6533	0.5936	0.4581	0.7187
22							
23	asw						
24	HumanBrainCellAtlas	0.5619	0.5967	0.5967	0.3116	0.3116	0.6813
25	TabulaSapiens_All_Cells_v1	0.5282	0.5365	0.5363	0.1730	0.1730	0.4764
26	TabulaSapiens_All_Cells_v2	0.4559	0.5341	0.4019	0.1836	0.1836	0.5045
27	PBMC 10k	0.6851	0.5526	0.5526	0.5742	0.5742	0.5862
28	average	0.5578	0.5550	0.5219	0.3106	0.3106	0.5621
29							
30	graph_conn						
31	HumanBrainCellAtlas	0.9942	0.9817	0.9817	0.4818	0.4818	0.9971
32	TabulaSapiens_All_Cells_v1	0.4738	0.4738	0.8128	0.4738	0.3697	0.7865
33	TabulaSapiens_All_Cells_v2	0.5483	0.5483	0.3856	0.5483	0.3401	0.8254
34	PBMC 10k	0.8987	0.9270	0.9270	0.9099	0.9099	0.9344
35	average	0.7288	0.7327	0.7768	0.6035	0.5254	0.8859

Correspondingly, the text will be updated from: “As demonstrated in Supplementary Fig. S19, CellFM achieved the highest average AvgBio scores on these datasets, outperforming the second-best method, UCE, by 2.1%.”

to: “As demonstrated in Supplementary Fig. S19, CellFM achieved the highest average AvgBio scores on these datasets, outperforming the second-best method, scGPT, by **2.9%**.”